Global Firms, Local Students
Multinational Presence Shapes College Major Choice

Job Market Paper

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Abstract

Multinational firms reshape local labor demand and the returns to skill acquisition. I document an additional channel through which they shape human capital formation: students choose different fields of study before entering the labor market. Combining administrative records from a large public university in Costa Rica with firm-level data on free trade zone establishments, I construct a measure of local multinational presence that weights firms by their distance to each applicant's district of residence and by how intensively each industry hires from each field of study. Estimating a conditional logit with district and year fixed effects and district-clustered standard errors, I find that a one standard deviation increase in within-district manufacturing exposure raises the probability of choosing STEM by 2.0 percentage points, five percent of the baseline rate, drawing students from social sciences and professional studies. Service-sector presence moves students in the opposite direction, a pattern difficult to reconcile with exposure proxying for local economic development. Women respond 62 percent more strongly than men to manufacturing presence, narrowing the 18 percentage point gender gap in STEM enrollment. The composition of inward FDI, not only its volume, shapes the skills a country's graduates acquire.

1 Introduction

Over the past several decades, foreign direct investments (FDI) have become a driving force in shaping developing economies (Koepke, 2019)[1]. In FDI-receiving countries, multinational corporations (MNCs) not only influence wages, productivity, and technological adoption[2], but also alter the composition of local labor markets (Dustmann and Glitz, 2015; Jude and Silaghi, 2016; Hale and Xu, 2016). A key, but under-explored question is how FDI may influence the skills that individuals choose to acquire through tertiary education (Blomström and Kokko, 2003; Meyer, 2004; Paul and Feliciano-Cestero, 2021). The introduction of MNCs into local economies shift employment opportunities toward two possible ends, low-skill or high-skill jobs. Labor supply, now driven by MNC demand, shifts toward new equilibriums (Blanchard and Olney, 2017). Employment trends have crowded-out middle-skill opportunities in industrialized economies (Goos and Manning, 2007; Goos et al., 2014) and we should expect similar shifts to follow in developing nations as MNC offer low-skill jobs (Feenstra and Hanson, 1997; Atkin, 2016) and high skill jobs (Javorcik, 2015; Alfaro Ureña et al., 2021; Setzler and Tintelnot, 2021).

This paper shows that MNCs influence labor markets by shifting students' demand for fields of study, which in turn reshapes the composition of the future high-skill labor force. I study this in Costa Rica, which has experienced significant growth in FDI inflows since 2009. This setting is particularly well-suited because of the many incentives that the country has provided to attract FDI and the detailed administrative level education data from a large public university. Further, Alfaro Ureña et al. (2021) have provided evidence of an MNC wage premium of up to 9% and a higher demand for highly educated individuals than domestic private firms in Costa Rica. By linking administrative education data with spatial variation in local MNC exposure over time, I examine how multinational presence shapes field of study

[1]FDI reached $916 billion (more than 70% of global flows, in 2022. $208 billion alone entered Latin America and the Caribbean) Giroud (2024)

[2]See Lipsey (2004); Jung and Mercenier (2009); Dustmann and Glitz (2015) for work on different channels and absorption of FDI

choices among prospective high-skill individuals.

Conceptually, prospective university students view a field of study as targeted skill acquisition which assists them in efforts to be employed in a given industry. For example, to obtain employment in the semiconductor manufacturing industry, they are generally aware that some form of engineering discipline would endow them with the necessary skills. Additionally, students are exposed to local labor market opportunities, which inherently vary by location. As FDI projects increase in quantity and given the greater attractiveness of MNCs relative to domestic private firms, existing foreign firms become more relevant to the student's decision-making process. MNCs may offer additional factors that make them attractive as they may allow for non-pecuniary returns, better future employment opportunities, or possible migration opportunities. As labor supply adjusts to a changing labor market composition, students adjust their skill acquisition decisions accordingly. This adjustment should be considered similarly to adjustments observed after trade shocks (Artuç et al., 2010).

To conduct an empirical analysis, I need data on student's demand for majors (before enrollment equilibrium) and MNC industrial composition and location to identify local labor markets. For data on student demand for major, I use administrative data on university applications to a large public university from 2010 to 2019. For data on MNC firms, I use firms that operate within the MNC-exclusive Free Trade Zone regime in Costa Rica. I then estimate MNC effects on field of study choice using a Multinomial Logit model.

I model field of study choice directly, which allows me to recover both the magnitude and the direction of these effects. Students are differentially exposed to multinational firms according to their district of residence and year of application. Exposure is captured by a presence index that weights each free trade zone firm by a Gaussian function of its distance to the applicant's district centroid, with a bandwidth of 35 kilometers. I then construct a matrix of field-to-industry attachment probabilities, measuring how intensively each industry hires from each

field, and interact it with the presence index. Exposure to a given industry therefore shifts the attractiveness of each field in proportion to that industry's observed demand for its graduates, which is what permits responses to differ across field-industry pairings rather than reflecting a uniform shift in the attractiveness of applying. Identification rests on variation in exposure to multinational firms across districts and years. District and year fixed effects absorb fixed differences across districts in the propensity to choose particular fields, along with national shocks common to all applicants in a year; because the fixed effects are alternative-specific, they absorb a free parameter for every district-field and year-field cell. What remains is within-district variation over time, generated by firm entry and exit changing the set of establishments near a given district. Standard errors are clustered at the district level, since exposure is assigned at district-year rather than to individual students.

The identifying assumption is that, conditional on individual characteristics and local economic conditions, changes over time in the industrial composition of nearby multinational activity are unrelated to unobserved determinants of field choice.

The structure of the results speaks to this concern. Exposure measures for different industries are highly correlated with one another, since multinational activity concentrates in the same metropolitan core regardless of sector. A confounder operating through general local development would therefore bias all industry coefficients in the same direction. The estimates instead show manufacturing and professional services moving students in opposite directions by comparable magnitudes, a pattern that requires something distinguishing between industries whose geography is nearly identical. I find that exposure to multinational presence shifts the probability that students select a given field, with effects differing across field-industry pairings. Manufacturing, the largest multinational industry in Costa Rica, draws students toward technical fields: a one standard deviation increase in within-district manufacturing exposure raises the probability of selecting a major within STEM and Applied Sciences by 2.0 percentage points, five percent of the baseline rate. This increase is drawn

four-fifths from Social Sciences and Professional Studies (1.6 percentage points) and one-fifth from Arts, Writing and Culture (0.4 percentage points). Professional, scientific and technical services move students in the opposite direction, lowering the probability of choosing STEM by 2.0 percentage points and raising Social Sciences and Professional Studies by the same order. Results are robust to an alternative index that weights firms by accumulated years of operation rather than by count.

Responses differ systematically by sex. Women's response to manufacturing presence is 62 percent larger than men's, while their response to professional services is 42 percent smaller. Because both industries therefore move women toward the STEM field relative to men, each narrows the existing 17.9 percentage point gender gap in STEM enrollment, although by amounts that shift the gap at the margin rather than closing it. That the differences run in opposite directions across the two industries argues against a general account in which women's choices are simply more responsive to local conditions. This paper contributes to three main strands of existing literature: (1) interactions between trade and human capital (2) impact of FDI on host economies and (3) determinants of major choice. By identifying and measuring the effects of MNC presence on field of study choices, I help explain a new mechanism through which trade impacts human capital decisions: the presence of multinational firms impacting individual skill specialization decisions. I also add to the growing literature that analyzes the effects of FDI-receiving nations. I demonstrate that FDI's impact on educational attainment extends beyond simple participation, shaping skill acquisition at the intensive margin as reflected in students' decisions regarding their field of study.

The paper is structured as follows. Section 2 describes the relevant literature and highlights contributions. Section 3 describes the institutional details and context of the project in greater detail. Section 4 outlines the conceptual framework behind the college major choice mechanism. Section 5 presents the data. Section 6 describes the empirical approach. Section

7 presents the empirical results and Section 8 concludes.

2 Literature Review

The relationship between international trade and human capital formation has been extensively studied in the endogenous growth literature. Foundational models, such as Romer (1990) and Caballé and Santos (1993), formalize human capital investment decisions as drivers of economic growth. As greater FDI volume enters developing nations, absorptive capacities of receiving nations may dictate efficient productivity gains (Jung, 2020). FDI has complementarities with schooling, where economic growth benefits may depend on existing levels of schooling. Wang and Wong (2011) show that these levels do not need to be necessarily high, with quality being an important determinant over quantity, as was previously shown in Borensztein et al. (1998). From a macroeconomic perspective, FDI has an impact on national average years of schooling attained (Blanchard and Olney, 2017; Kheng et al., 2017; Dey and Mishra, 2018), and aggregate enrollment counts (Mughal and Vechiu, 2015). In OECD countries, tertiary education has a great influence in attracting inward FDI (Pantelopoulos, 2022). Crucially, Blanchard and Olney (2017) provide evidence that these effects are likely determined by the type of sectoral growth: skill-intensive exports increase schooling.

Looking at empirical evidence, FDI impacts receiving economies through backward linkages between multinationals and domestic firms (Smarzynska Javorcik, 2004), skill upgrading of workers (Jung and Mercenier, 2009), wage premia (Javorcik, 2015; Alfaro Ureña et al., 2021), technology transfers (Zhuang, 2017), and labor market composition changes (Blanchard and Willmann, 2016). Importantly to the context of this paper, Alvarado et al. (2017) cautions on the efficacy of FDI in driving growth in Latin America, where high-income countries seem to be the only true beneficiaries.

FDI inflows may also contribute to broader structural transformations of local markets,

through industry composition and patterns of skill demand in the host economy. Blanchard and Willmann (2016) develop a model of skill acquisition where trade impacts education and employment attainment. They conclude that targeted educational policies would benefit middle-skilled workers when exposed to effects of increased trade. Empirically, Goos and Manning (2007) and Goos et al. (2014) highlight a polarization of jobs toward the skill-distribution extremes; low- and high-skill jobs. There is evidence of negative effects on educational attainment levels stemming from greater trade (Feenstra and Hanson, 1997; Atkin, 2016) and demand for higher-skill labor (Javorcik, 2015; Setzler and Tintelnot, 2021).

This paper also relates to literature of determinants of major choice. A large proportion of existing work has focused on the impact of expected earnings on major choice. Observational studies (Altonji, 1993; Carneiro et al., 2011; Beffy et al., 2012; Hastings et al., 2013; Kirkeboen et al., 2016) and laboratory experiments (Arcidiacono et al., 2012, 2020) suggest that expected earnings are an important determinant of university major choices.[3] Since students are directly influenced by future expected earnings, the wage premia offered by multinational firms should be a direct influence on field of study choices by students. Availability of information is also an important factor for students' decision-making. Previous research shows that changing expectations of returns to education lead to changes in educational attainment efforts. In work done by Jensen (2010), eighth-grade boys in the Dominican Republic completed 0.20 to 0.35 more years of school, on average, after learning of higher returns to education.

Another important factor influencing individuals' choice of field of study are their endogenous abilities, a phenomenon often referred to as ability sorting. Arcidiacono (2004) confirms that an earning premia exist as a determinant of choice but are not able to fully explain sorting across majors. They state that sorting is mainly due to individual preferences for a given field. Kinsler and Pavan (2015) provide evidence that selection into majors plays a role in generating wage gaps across fields. Intuitively, preferences are subject to their surrounding

[3]See Altonji et al. (2012) for a review of both theoretical and empirical existing literature.

environment and are subject to change. In this project, individuals are subject to their local labor markets and shifting MNC compositions.

The existing literature contextualizes the choice of major through expected wages and endogenous abilities. This paper contributes to the determinants of major choice by highlighting a different channel: the influence of their surrounding labor market composition. I study how the expansion of multinational firms in developing economies shapes students' field of study decisions through changes in the local labor market industry composition of MNCs.

3 Institutional Details

Costa Rica has a total of four public universities: Universidad de Costa Rica (UCR), Universidad Nacional (UNA), Instituto Tecnológico de Costa Rica (TEC), and Universidad Técnica Nacional (UTN). Tuition costs are generally low in Costa Rica, with a university credit costing approximately $30 in 2025.[4] Obtaining a tertiary degree (I only consider bachelor's degrees in this project) will usually take students 5 years on average. Due to data availability, I only observe applications into UNA representing a share of 21% of university students.

Admissions Process

The admissions process to a public university in Costa Rica is conducted using a decentralized process, where individuals apply directly to each university separately. In order to apply to any public institution, the applicant must take an entry exam meant to evaluate verbal and mathematical reasoning skills. Universities generate an entry score for each prospective student through a weighted average of their performance on the entry exam and their performance over the last two years of their high school education. After receiving their

[4]This corresponds to nearly $400 for a full credit load semester. Over 50% of students receive some form of scholarship.

scores, each student is asked to submit their two preferred majors, in order.[5] The institution determines, through budgetary and internal processes, how many seats it will award for each major each year but this is unknown to the student at the time of application.

However, the institutions make public the historic cut-off grades for being admitted into each major. Although this does not guarantee admission simply by having a higher entry score than previous years, it does provide the student with important information as they choose majors. Once the application deadline passes, the university rank-orders each student by major and entry score. The university then admits students into their first choice major going down in order of entry score until there are no more seats available. This constitutes the "cut-off" grade for being admitted into the major, which is essentially the last student to have been accepted. Once this sorting finalizes, students are informed of their application status, and they are allowed to enroll into the institution.

Free Trade Zone Requirements & Incentives

The Free Trade Zone regime in Costa Rica is a set of incentives and benefits given to firms that realize new FDI projects that meet all requirements. It is mandated by Free Trade Zone Law N7210 and all articles therewithin. Firms that are accepted into the regime establish their activities within determined industrial parks which are strictly designated for firms operating under the FTZ regime. It is possible for firms to obtain authorization from the government to establish operations outside the industrial park but these are rare cases. For simplicity, I remove all firms that are reported as operating outside of the industrial parks in the data.[6]

There is a required minimum investment amount of $150,000 in fixed assets. This investment requirement is less for firms that locate themselves outside the Greater Metropolitan Area

[5] An example from Universidad de Costa Rica can be seen in figure **??**.

[6] Firms may request special permission to operate outside of the Industrial Park but this is subject to approval by the governing body.

(GAM, from the Spanish acronym), being required to a minimum investment amount of $100,000. These investments must be completed in the initial 3 years after the firm has been notified of their acceptance into the regime. Beyond the initial financial investment requirement, firms must also abide by both local and international strict environmental policies and must be able to prove they abide by them.

Firms are allowed to carry out all regular business activities, plus other specific ones outlined by the law, without limiting their overall scope. Firms that provide services can serve both firms in the FTZ and foreign clients. Manufacturing and processing firms can perform a broad range of activities on goods for export/re-export, except for activities prohibited by provisions in the law. Outsourcing of services is allowed without limitations if these services are provided to foreign firms or to firms also operating within the FTZ regime. If services are provided to firms outside the FTZs, only a maximum of 50% of total sales may come from these clients. Similarly, local sale of products by processing firms may represent up to 25% of the firm's total sales, except in the case that the firm is labeled as a "Type f processing firm", as indicated in subsection (f) of Article 17 of the FTZ Law. Subsection (f) firms may sell 100% of their output in the local market, meaning that they are not required to export in order to benefit from the FTZ regime.

4 Conceptual Framework

This section provides a stylized model that isolates the mechanism behind the empirical analysis below. There are two assumptions made. First, wages do not factor into the individual's utility. Expected pay is common across industries and absorbed into the field-specific intercepts, so that exposure can affect choices only through the non-wage channels (probability of finding work in an attached industry, information, and networks). Second, firm presence is taken as exogenous. Multinational firm entry and location are not endogenously determined.

4.1 Environment

Students are indexed by i, districts by d, years by t, fields of study by $m \in \{1, \ldots, M\}$, and industries by $k \in \{1, \ldots, K\}$. Each student chooses one field or an outside option $m = 0$. Multinational firm presence is given by $\Gamma_{kjt} \geq 0$ in industry k and district j is exogenous.

Distance Weights

The influence of a firm on students in district d depends on the distance $dist_{dj}$ (in kilometers) between the two districts through a Gaussian kernel,

$$\omega(dist_{dj}) = exp\left(-\frac{dist_{dj}^2}{2h^2}\right), \tag{1}$$

with bandwidth $h = 35$. The kernel equals one for firms in the student's own district and falls toward zero beyond a few bandwidths. Unlike an exponential, it is flat near the origin so tha tfirms within a short distance are weighted almost equally and the weight drops quickly once distance exceeds h. The bandwidth is set at the a rough estimation of the GAM diameter rather than estimated.

Exposure

Let $a_{mk} \geq 0$ with $\sum_m a_{mk} = 1$ be the attachment of industry k to field m, the share of industry k's demand for skilled labor that is absorbed by graduates of field m. Exposure in district d to industry k is

$$\Gamma_{mdt} = \sum_{k=1}^{K} a_{mk} \sum_{j} \omega(dist_{dj}) F_{kjt}. \tag{2}$$

The entry and continued existence of multinational firms in industry k raises exposure of every field, weighted by the attachment to that industry, and more so for students living closer to the firm's district of operation.

Preferences

The utility of student i from choosing field m is

$$U_{imdt} = \alpha_m + \beta \Gamma_{mdt} + \gamma'_m X_i + \varepsilon_{imdt}, \tag{3}$$

where α_m is a field intercept, X_i are individual characteristics entering through field-specific coefficients γ_m, and ε is i.i.i.d. type-I extreme value. The coefficient β summarizes all non-wage exposure channels through which exposure impacts the appeal of a field.

4.2 Choice Probabilities

Student i chooses field m with probability

$$P_{imdt} = \frac{exp(\alpha_m + \beta \Gamma_{mdt} + \gamma'_m X_i)}{1 + \sum_{l=1}^{M} exp(\alpha_l + \Gamma_{ldt} + \gamma'_l X_i)}, \tag{4}$$

and the outside option with probability $P_{i0dt} = 1 - \sum_m P_{imdt}$. Equation 4 is the conditional logit with alternative-varying exposure estimated in the empirical sections below, so the coefficeint on exposure in that specification identifies β.

4.3 Comparative Statics

Proposition 1 *Holding F fixed outside industry k and district j,*

$$\frac{\partial \Gamma_{mdt}}{\partial F_{jkt}} = a_{mk}\, \omega(dist_{dj}) \tag{5}$$

$$\frac{\partial P_{imdt}}{\partial \Gamma_{mdt}} = \beta P_{imdt}(1 - P_{imdt}) \tag{6}$$

$$\frac{\partial P_{ildt}}{\partial \Gamma_{mdt}} = -\beta P_{imdt} P_{ildt}, \quad l \neq m, l \in \{0, 1, \ldots, M\}. \tag{7}$$

Consequently, the total effect of an increase in industry-k presence in district j on the probability of field l is

$$\frac{\partial P_{ildt}}{\partial F_{jkt}} = \beta\, \omega(dist_{dj}) P_{ildt} \left(a_{lk} - \sum_{m=1}^{M} a_{mk} P_{imdt} \right) \tag{8}$$

for $l \in 1, \ldots, M$, and for the outside option, we have:

$$\frac{\partial P_{i0dt}}{\partial F_{kjt}} = -\beta\, \omega(dist_{dj}) P_{i0dt} \sum_{m=1}^{M} a_{mk} P_{imdt}. \tag{9}$$

Proof 1 *Equation (5) follows from differentiating equation (2). Equations (6) and (7) are the standard multinomial logit derivatives. For equation (8), sum eq. (5) against (6) and (7): Note that Γ_{mdt} moves the exposure of every field at once, then by the chain rule we can write:*

$$\frac{\partial P_{ildt}}{\partial F_{jkt}} = \sum_{m=1}^{M} \frac{\partial P_{ildt}}{\Gamma_{mdt}} \cdot \frac{\partial \Gamma_{dmt}}{\partial F_{kjt}}.$$

Separating the term $m = l$ from the terms $m \neq l$. The first term uses (5) and (6), the other term uses (5) and (7)

$$\frac{\partial P_{ildt}}{\partial F_{jkt}} = \beta P_{ildt}(1 - P_{ildt}) \cdot a_{lk}\omega(dist_{dj}) + \sum_{m \neq l} (-\beta P_{imdt} P_{ildt}) \cdot a_{mk}\omega(dist_{dj})$$

$$\frac{\partial P_{ildt}}{\partial F_{jkt}} = \beta\omega(dist_{d}j) P_{ildt} \left[a_{lk}(1 - P_{ildt}) - \sum_{m \neq l} a_{mk} P_{imdt} \right].$$

Expanding $a_{lk}(1 - P_{ildt}) = a_{lk} - a_{lk}P_{ildt}$ and combining $-a_{lk}P_{ildt}$ with the sum over $m \neq l$ gives $-(a_{lk}P_{ildt} + \sum_{m \neq l} a_{mk}P_{imdt}) = -\sum_{m=1}^{M} a_{mk}P_{imdt}$. Hence:

$$\frac{\partial P_{ildt}}{\partial F_{jkt}} = \beta\, \omega(dist_{dj}) P_{ildt} \left(a_{lk} - \sum_{m=1}^{M} a_{mk} P_{imdt} \right)$$

which is (8). The outside option case is analogous, using $a_{0k} = 0$.

Remark 1 *Equation (8) delivers three testable implications.*

1. *Field l gains if and only if its attachment to industry k exceeds the probability-weighted average attachment, $a_{lk} > \sum_m a_{mk} P_m$ (for $\beta > 0$); fields weakly attached to k lose.*

2. *The response is scaled by the Gaussian weight $w(\text{dist}_{dj})$, so it is nearly undiminished for firms within a fraction of h and decays rapidly beyond. With $h = 35$, a plant 50 km away carries a weight of about 0.36 and one 100 km away about 0.02.*

3. *Effects are proportional to the baseline probability P_{ildt}, so average marginal effects depend on where in the distribution the student sits, which is why marginal effects must be evaluated within the support of the data. Because $\sum_{l=0}^{M} \partial P_l / \partial F_{kj} = 0$, a rise in exposure reallocates students across fields and the outside option rather than creating them.*

5 Data

MNCs Operating Within FTZ Regime

Firms under this regime are by definition Multinational Corporations. I begin with the entire universe of firms operating within the FTZ regime as of 2023. Data is sourced from the Costa Rican Foreign Trade Promoter (PROCOMER, for its name in spanish). The data contains each firm's name, location at the province-canton-district[7] level, whether they operate inside the industrial park or not, industries they engage in, and a unique corporate identity number. A crucial part of my identification strategy is the time of entry of each firm, which is missing in this data. To remedy this, I use each firm's unique corporate identity number to identify their earliest date of registry with the Costa Rican government using the Costa Rica National Registry.[8] From this data I select the earliest official document's date possible and assign it to the respective firm.

The economic activity codes are equivalent to the 4-digit International Standard Industrial

[7] Costa Rica is administratively organized into 7 Provinces, 81 Cantons, and 488 Districts.
[8] Access to the registry can be completed online at http://www.registronacional.go.cr

Classification (ISIC Rev.4). For simplicity, I use the first reported code in the data. I then concord these codes to their 2-digit categories in order to properly map it with the industry by field of study mapping I detail below. This produces 31 unique 2-digit ISIC Rev. 4 categories, with the largest proportion of firms being in Manufacturing (29.5%). The full list can be found in table A1 of the Appendix. Figure A2a and Figure A2b show the aggregated spatial distriution of the firms.

Education

Data on university applications from the Universidad Nacional (UNA) are sourced directly from the university. Each observation is an anonymized individual application containing the year they applied, their province-canton-district of residence, age, sex, type of high school attended, entry score grade, their first and second choices for major and the respective International Standard Classification of Education (ISCED) category as determined by UNESCO. For UNA, there are 77 unique majors which are categorized into 10 Broad Categories. To make the possible choice set for each student more tractable, I aggregate the possible majors using the ISCED broad categories into three fields-of-study.[9] These are Arts, Writing and Tourism, STEM and Applied Sciences, and Social Sciences and Professional Studies. These are the possible choices each individual will face when completing their application. Table A2 in the Appendix shows the categories with examples of majors within each category.

The data from UNA expands from 2010 to 2019 and contains 230,162 applications. The sample leans more female (57%), an average age of 19.49 years, and over 75% of applicants come from public high schools. The majority field of study chosen are disciplines on Social Science and Professional Studies (50.7%) and STEM being the second largest (36%). Application by regions logically follow population patterns, where most come from the largest provinces. Figure A3a and Figure A3b show the aggregated spatial distriution of the firms.

[9] A diagram showing the workflow can be found in figure A1 in the Appendix.

In Section 6, I detail how I map industries to the fields of study in my analysis. They are probability attachment weights which capture how likely a given field of study is to have employment in a given industry. To do so, I use the American Community Survey (ACS) for each year of the sample period.[10] The ACS surveys individuals on their education attainment and which industry they are employed in, if any, amongst many other questions. One question asks respondents which field they have a degree in, which using the same ISCED categorization structure as in the education data above, I sort into the three aggregated fields of study.

6 Methodology

Model

In order to estimate how MNC presence impacts field of study choice, I model individuals utility as:

$$U_{idmt} = \beta_m \Gamma_{dmjt} + \gamma_i X_i' + \alpha_d + \alpha_t + \varepsilon_{idmt}, \tag{10}$$

where individual i from district d chooses the field of study m in year t that maximizes their utility. β_m captures the effect of MNC presence on field of study choice. The MNC industry presence index (Γ_{djt}) measures the level of influence that any given industry has on the student. The index itself is weighed by the probability attachment weights. In this way, fields with a higher likelihood of being employed in a given industry receive a more appropriate amount of influence, and vice-versa. Each individual also has demographic characteristics X_i', like their age, sex, entry score, high school type and the local unemployment rate. They are also subject to canton and year fixed effects that absorb unobserved factors that will influence all individuals in the same canton or year. While the fixed effects shift the overall level of utility, they do not affect choice probabilities directly in the multinomial logit

[10] I am unaware of similar data existing for Costa Rica.

regression as the choice depends on differences on utility across alternatives. Individuals are also subject to an idiosyncratic shock ε_{idmt}. The possible field of study choices are those noted in table **??**.

The individual choice is estimated using a Multinomial Logit (MNL) model where individuals choose the field of study which gives them the highest utility:

$$P(Y_i = m) = \frac{\overbrace{exp(\beta_m \Gamma_{dmjt} + \gamma_i X_i' + \alpha_d + \alpha_t + \varepsilon_{idmt})}^{\text{Utility of chosen field of study}}}{\underbrace{\sum_{m'} exp(\beta_{m'} \Gamma_{dm'jt} + \gamma_i X_i' + \alpha_d + \alpha_t + \varepsilon_{idm't})}_{\text{Utilities across all possible field of study choices}}} \tag{11}$$

This gives the probabilities of choosing field of study m as determined by the utility function above. The coefficients of interest are the β_m, which capture the effects of MNEs through their industry to choose a field of study m. Specifically, Γ_{dmjt} is an index of the presence of an industry j in year t weighed by the distance between firms and individual i. This ratio is then multiplied by the attachment probabilities between field of study m to industry j for each year t. These coefficients quantify the extent to which students' field of study choices respond to influences of MNCs in the local labor market.

MNC Presence Index

In order to create an index that captures how individuals experience the presence of MNCs, I use the following gravity model structure:

$$z_{djt}(h) = \sum_{d'} N_{jd't} \cdot exp\left(\frac{dist_{dd'}^2}{2h^2}\right), \quad h = 35km$$

$$\Gamma_{dmjt} = p_{mjt} \cdot z_{djt}(h), \tag{12}$$

where $N_{md't}$ is either the count or tenure of industry-j firms located in district d' at year t, $dist_{dd'}$ is the centroid-to-centroid distance in kilometers, and p_{mjt} are the probability

attachment weights between industry j and field m in year t.

Given how the MNC presence index is constructed, an increase of one unit in the index can potentially arise from two events: (1) individuals relocate closer to the firms or (2) new firms enter the economy. Because the data are repeated cross-sections, there is no possibility of migration. Without migration as a mechanism, this leaves new firms entering or aging as the only possibility of an increase. Given that the index is also weighed by spatial distance, one new firm does not necessarily equal one unit increase in the index. An increase of one unit should be interpreted as an additional firm-year in the student's district of residence. For increased clarity, the results are interpreted using changes in intensity of the index. These are shifts from low- to high-presence of industries for the individual.

Average Marginal Effects

To better interpret the effects of MNCs on prospective students' field of study choices, I compute the average marginal effects (AMEs) from the above model. While the multinomial logit model coefficients β_m describe how industry-specific presence Γ_{dmjt} affects the latent utility of choosing a specific field of study m, they are not directly interpretable in terms of choice probabilities due to the non-linear form of the logit model.

AMEs allow for a direct interpretation by transforming the estimated logit utilities into changes in predicted probabilities. Specifically, the marginal effect of an industry presence variable Γ_{djt} on the probability that individual i chooses field of study m is given by:

$$\frac{\partial P_{ijt}}{\partial z_{djt}} = \beta_m \, P_{ijt} \left(p_{mjt} - \sum_k P_{ikt} \, p_{mkt} \right) \tag{13}$$

where P_{im} is the predicted probability that individual i chooses field of study m, β_j is the coefficient on exposure to industry j, and p_{mjt} is the share of industry j's hiring drawn from field m. The summation is the probability-weighted average attachment weight across the

student's three alternatives, so the term in parentheses measures how much more intensively industry j hires from field m than from the student's choice set as a whole. A field gains probability only if its attachment to the expanding industry exceeds that average. Because the deviations are weighted by the same probabilities, the marginal effects sum to zero across fields: the expression describes a reallocation of probability mass across competing alternatives, not a change in the level of participation.

I compute the average marginal effects, which essentially computes this for each individual and averages over the sample:

$$\text{AME}_{mj} = \frac{1}{N} \sum_{i=1}^{N} \frac{\partial P_{im}}{\partial z_{\,d(i)\,j\,t(i)}} \tag{14}$$

These AMEs represent the expected change in the probability of choosing field of study m with a one-unit increase in the presence index of industry j, holding all else equal. Because these are direct predictions on changes in probability, there is no reference category and all probabilities should sum to zero. The coefficients shown in all regression tables are in percentage points, with a positive value indicating an increase and a negative value a decrease in the probability of that field of study being chosen.

Education to Industry Mapping

A crucial detail for this project is the ability to map individual industry effects to possible fields-of-study. Given that any university degree can potentially find employment across multiple industries, it is necessary to address how I can differentiate across education field by industry pairings. Using the American Community Survey, I leverage two crucial questions asked. Each individual is asked if they have a degree and if so, in which field. Additionally, they are asked if they are employed and if they answer yes, then they are assigned an industry. I concord the degree/field question using the same ISCED categorization as is used for the university application data. Reported industries of employment use the North American

Industry Classification System (NAICS) with varying amount of detail by individual. I concord them to their 2-digit classifications and then concord those to their ISIC Rev.4 equivalents so they match up with the MNC firm data. This creates a harmonized dataset, which allows me to create a probabilistic attachment weight from each field of study to observed industries.

These attachment weights allow me to assign probabilistic relevance of each field of study to each existing industry. This reflects the fact that, although any given industry, for example, may hire an individual with a degree in chemical engineering they may also find employment with a financial firm although that is less likely. It creates differential effects across the education by industry pairings choice set. The mapping can be found in table A3 and the within-sample distribution of the weights can be found in figure A4.

7 Results

I begin with estimating how local industry presence shifts the field a student chooses, conditional on applying. Estimates are reported as average marginal effects in percentage points per one within-district standard deviation of the exposure index and because the three fields partition the choice set, they describe a reallocation of probability across fields rather than a change in its level.

To estimate the robustness of these results, I then re-estimate the main specification removing the probaiblity attachment weights between industry and field of study. Lastly, I run a heterogeneity test on whether responses vary by sex.

7.1 Average Marginal Effects of Multinational Presence on Field of Study Choice

I now turn to how local multinational presence shapes the composition of applicants across fields of study. These are intensive-margin effects as they describe which field a prospective student selects, conditional on applying. Table A4 and table A5 report the average marginal effects of industry presence on the probability of choosing each of the three field categories.

Each estimate is expressed in percentage points per one within-district standard deviation increase in the respective industry exposure index. This scaling is not comestic. Because the specification absorbs district and year effects, identification comes from changes in nearby firm presence over time within a district. Reporting effects per raw index unit, or per standard deviation of the unconditional distribution, would describe variation the estimator does not use. For clarity, one within-district standard deviation of manufacturing exposure corresponds to 2.8 index units, or approximately three additional firms (when considering count) or 3 additional firm years (when considering firm tenure) in a student's own district. Standard errors are clustered at the district level, since exposure is assigned at district-year rather than to the individual student. Because the three categories partition the choice set, marginal effect sum to zero within each industry by construction. A gain in one field is necessarily drawn from the others, thus the three estimates for a given industry sum to zero. The industry-specific estimates are proportional to a single coefficient and share one test statistic, so the table reports seven independent tests rather than 21 tests.

Manufacturing is the natural industry on which to focus. Over the sample period it represents the largest share of FDI inflows (36.8 percent), ranging between 0.67 and 3.05 percent of GDP. It is also the only industry in Tables A4 and A5 that draws students toward technical fields. A one standard deviation increase in within-district manufacturing exposure raises the probability of choosing a major within the STEM and Applied Sciences field by 2.02 percentage points. This represents a five percent increase relative to the baseline STEM

share of 40.1 percent. The multinomial structure identifies where these students come from: roughly 80 percent of the increase is drawn from Social Sciences and Professional Studies (-1.61 p.p.) and 20 percent from the Arts, Writing, and Culture category (-0.41 p.p.). Measured against each category's own baseline, however, the two losing fields are comparably affected, with declines of 3.2 and 4.5 percent respectively. Multinational firms presence in the manufacturing industry therefore operates primarily on the STEM-Professional studies margin, while bearing proportionally just as heavily on the smaller Arts category.

Turning to the Professional, Scientific, and Technical Activities industry, the results suggest that an increase in the presence of this industry moves students in the opposite direction. A one standard deviation increase in exposure to this industry lowers the probability of choosing a major within the STEM field by 2.05 percentage points and raises the probability of choosing Social Sciences and Professional Studies by 1.97 points. The utilities and wholesale trade display the same orientation, though at roughly a third of the magnitude. The constrast is consistent with the occupational composition of these sectors, where it is more likely that professional activities hire disproportionately more accounting, legal, and management fields, while manufacturing hires engineers and technicians.

This pattern also bears on identification. If the exposure measure were principally capturing local economic development, we would expect to see that districts with higher concentration of multinational firms to carry the same coefficient sign, differing only in magnitude. Instead, manufacturing and professional activities generate effects of comparable size and opposite directon. A common unobserved factor cannot produce that pattern. It requires somehting that distinguises between industries whose geography is almost identical, which is precisely what the field-specific probability attachment weights do. Administrative support services and information services are statistically indistinguishable from zero in both directions, further suggesting that measured exposure is not simply tracking the presence of firms as such.

Two other estimates warrant qualification. Transportation and warehousing is precisely estimated but economically negligible. Wholesale trade sits at the margin of conventional significance. Neither should be read as a substantive finding.

Responses also differ by sex. Women respond to manufacturing exposure by 62 percent more than men do, while their response to professional activities is 42 percent smaller. Because both industries move women toward STEM relative to men, each narrows the 17.9 percentage point gender gap in STEM enrollment, by 0.67 and 1.36 percentage points per standard deviation respectively. These differences are estimated jointly with the levels and are discussed in Section 7.3.1 below.

7.2 Robustness Tests

7.2.1 No Attachment Probabilities

The main specification imposes a mapping between industries and field of study through the attachment weights p_{mjt}, which captures how intensively each industry hires from each field. A natural test would be whether the results depend on that imposed structure. I re-estimate the model allowing the data to determine the mapping.

To do this, it first requires a slight change in the specification rather than simply dropping the weights. Without the weights, the exposure index takes the same value for all three of the student's alternatives, and these would cancel out from the choice probabilities entirely. The index must enter as a case-specific regressor, so that a separate coefficient is produced for each non-reference category. Each industry then shifts the relative attractiveness of each field freely, with no restriction linking those shifts to observed industry-field patterns. The cost is that the specification estimates 14 exposure paratmeters rather than seven. The unweighted AMEs are then estimated by:

$$\frac{\partial P_{im}}{\partial z_{djt}} = P_{im}\left(\beta_{mj} - \sum_{m'} P_{im'}\,\beta_{m'j}\right) \tag{15}$$

Table A6 reports the results. The manufacturing effect is unchanged in direction and remains statistically significant, with an AME estiamte of 3.83 percentage points against 2.02 in the main analysis. Professional, scientific, and technical activities retains its negative sign on STEM but the magnitude is halved and loses its statistical significance. Looking at other industries, utilities and wholesale trade, reproduce almost exactly (-0.71 versus -0.70 and -0.52 versus -0.50, respectively). Every industry that was significant in the main model preserves their direction on its effect on STEM choice.

What differs between the two specifications is how the effects are attributed across industries. The summed effect on the probability of choosing STEM across industries is -0.86 percentage points in the main results and -0.71 in the unweighted specification. By lifting imposed restrictions on the model, the industries trade off against one another without altering their combined contribution. The attachment weights discipline this process, by more accurately resolving which industry receives credit for the effects.

Two further caveats apply to this analysis. Because each industry now has two freely estimated coefficients rather than one, the three category0specific effects within an industry are no longer proportional to a common parameter and no longer share a test statistic. And because the two specifications are non-nested, they cannot be compared by a likelihood ratio test.

Taken together, the exercice supports the main result while helping qualify the industry-level decomposition. The finding that manufacturing presence draws students toward STEM fields, and that non-manufacturing presence does the opposite, does not depend on the attachment weights.

7.3 Heterogeneity Test

7.3.1 Differences by Sex

Field of study is among the most gender-segregated margins in higher education, and the baseline figures in the sample follow suit: 50.4 percent of men choose a major within the STEM and Applied Sciences versus 32.5 percent of women, a gap of 17.9 percentage points. If multinational firm presence operates by making particular industries and occupations visible and their returns salient, there should be no reason to expect men and women to respond identically. Any existing difference would have FDI either widen or narrow this gap.

I estimate the heterogeneity between men and women by interacting each industry's exposure measure with an indicator for female within a single pooled specification. This modeling choice matters as coefficients in discrete choice models are identified only up to scale. Comparing coefficients across separately fitted models would confound the differences in the effect with differences in residual variances and the two cannot be separated (Allison, 1999; Mood, 2010). A pooled specification imposes a common scale by construction, so the interaction coefficient is a clean test of a difference between the categories. Due to the significance estimated above of manufacturing and professional, scientific, and technical activities industries, I only interact the female indicator with those two industries. The remaining industris enter without interactions, and the specification is otherwise identical to the baseline.

Results of this analysis suggest that women respond more strongly than men to manufacturing presence. The estimated response coefficient for women is 1.62 times than for men, implying a response 62 percent larger. In marginal effects, a one standard deviation increase in the within-district manufacturing exposure raises the probability of choosing STEM by 1.63 percentage points for men and 2.31 points for women, a difference of 0.67 points which is statistically significant.

This pattern reverses for professional, scientific, and technical activities. Here, the women's response coefficient is 0.58 times the men's, a 42 percent smaller response. Exposure to this

industry lowers the probability of choosing STEM by 2.78 percentage points for men and 1.42 points for women, a statistically significant difference of 1.36 percentage points. The reversal between industries is informative. If it were the case that women's field choices were simply more sensitive to local MNC conditions would require response difference of the same sign across industries. Instead the differences are industry-specific. Women are pulled more strongly toward STEM by manufacturing and pushed less strongly away from it by professional activities. Both move women toward the field of STEM and applied sciences relative to men, but through opposite routes.

Interpreting the marginal effects requires one qualification. Marginal effects scale with the baseline choice probability, and women's baseline STEM share is 17.9 points below men's. For manufacturing, which raises the STEM probability, this smaller baseline works against the observed difference. Women show a larger marginal effect despite a mechanical force pushing it down, which suggests that the underlying behavioral difference is larger than the marginal effects coefficient suggest. For the professional activities industry, which lowers the STEM probability, the same forces work in the opposite direction and the marginal effect difference overstates the behavioral gap. The response ratios reported above are free of this composition channel and are the appropriate basis for the behavioral claim. The marginal effects describe what actually happens to the choice probabilities.

Taken together, both industries narrow the gender gap in STEM enrollment, by 0.67 and 1.36 percentage points per standard deviation respectively, or 3.8 and 7.6 percent of the baseline gap. The direction is consistent and the effects are precisely estimated, but the magnitudes should be treated with caution. To fully close the gap through manufacturing exposure alone would require roughly 27 standard deviations of additional presence, a quantity far beyond reality and the support of the data. Industry presence meaningfully shifts the gap at the margin, but it does not come close to eliminating it.

Table A7 and Figure A7 presents these estimates. In Figure A7, the left panels show the

marginal effects by sex and the right panels the female-minus-male difference. The subgroup confidence intervals ni the left panels overlap substantially, which might appear to contradict the precision of the differences, but they do not. Both subgroup effects are functions of the same exposure coefficient and are therefore strongly positively correlated ($\hat{\rho} = 0.99$). Uncertainty about the overall magnitude of the effect moves them together and cancels in their difference. Overlap between marginal confidence intervals carries no implication for the significance of a difference, particularly when the estimates are dependent (Schenker and Gentleman, 2001). Figure A8 makes this explicit. Across the 1,500 simulation draws underlying the confidence intervals, not one draw yields a female response below the male response.

One reading of the pattern is that women's field allocation is less entrenched at baseline. With half of men already choosing a major within STEM, fewer are near the margin of indifference. With only a third of women choosing STEM, more are considered marginal in their field of study decision, where a change in local labor demand can move them. The same logic operates in reverse for the push away from STEM, where there are more men available to be drawn away. The data identify the pattern rather than the mechanism behind it, and this interpretation should be treated as suggestive.

8 Conclusion

Human capital attainment is an important determinant of economic growth, and foreign direct investment havs become a common instrument of growth stragegy in developing economies. Educational attainment decisions are influenced by many factors, among them expected wages and endogenous ability sorting. This paper argues that the composition of local labor demand is an important factor to consider.

I show that foreign direct investment influences educational investment at the intensive margin, shaping the types of skills students acquire through their choice of field when applying to

university. Using administrative records on university applicants and spatial and temporal variation in exposure to multinational firms, I find that a one standard deviation increase in within-district manufacturing presence raises the probability of choosing STEM and Applied Sciences by 2.0 percentage points, drawing 80 percent from Social Sciences and Professional Studies. The effect survives district and year fixed effects, district-level clustering, and an alternative construction of the exposure index that weights firms by accumulated years of operation rather than by count.

The central insight is that which industries arrive matters, not only how many firms do. Manufacturing is the sole industry that draws students toward technical fields, while professional activities, utilities, and wholesale trade move them in the opposite direction, toward social sciences and professional programs. Because the industry exposure measures are highly correlated with one another, via multinational activity agglomerating in the greater metropolitan area, these opposing signs cannot be generated by a common unobserved factor such as local economic development. They require something that distinguishes between industries whose geography is nearly identical, which is why I employ field of study-specific hiring composition probabilities. Inward investment of different composition produces different human capital, even at the same volume.

The central result does not depend on the field-industry attachment weights. Allowing the data to estimate the mapping between industries and fields, rather than imposing it from observed hiring patterns, leaves the manufacturing effect on STEM choice positive, significant and somewhat larger, while utilities and wholesale trade reproduce almost exactly. What the weights affect is attribution rather than aggregate: the combined effect of industry exposure on STEM choice is essentially unchanged, while its allocation across non-manufacturing industries shifts. The direction of the manufacturing result is therefore robust to the imposed structure; the industry-level decomposition should be read with more caution.

Responses differ systematically by sex, and in a direction that narrows existing disparities.

Women response to manufacturing presence is 62 percent larger than men's, while their response to the professional activities industry is 42 percent smaller. Because both industries move women toward STEM fields relative to men, they both narrow the 17.9 percentage point gender gap in STEM enrollment, though by magnitudes that shift the gap at the margin rather than fully closing it. That the differences are industry-specific, rather than uniformly larger for women, argues against a general thought that women's choices in field of study are simply more sensitive to local conditions. It rather provides evidence of there being a larger share of women near the margin of indifference between fields.

Taken together, the evidence indicates that foreign direct investment shapes the future skill composition of the domestic workforce by influencing educational investment before labor market entry. This is a channel distinct from short-run employment effects, and it operates on a skill acquisition margin, what prospective students choose to study, that is slow to reverse. For policymakers using FDI as a development instrument, it suggests that the sectoral composition of what is attracted deserves at least as much attention as the volume.

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Appendix

Appendix A - Tables

Table A1: Industry Counts by ISIC-4 (2-Digit) Code

2-Digit Code	Industry	Count
	Manufacturing (103 Firms)	
10	Manufacture of Food Products	20
11	Manufacture of Beverages	2
13	Manufacture of Textiles	2
16	Manufacture of Wood and of Products of Wood and Cork, except Furniture	2
18	Printing and Reproduction of Recorded Media	1
20	Manufacture of Chemicals and Chemical Products	10
21	Manufacture of Basic Pharmaceutical Products and Pharmaceutical Preparations	2
22	Manufacture of Rubber and Plastics Products	8
23	Manufacture of Other Non-Metallic Mineral Products	2
24	Manufacture of Basic Metals	2
25	Manufacture of Fabricated Metal Products, except Machinery and Equipment	7
26	Manufacture of Computer, Electronic, and Optical Products	5
27	Manufacture of Electrical Equipment	2
32	Other Manufacturing	33
33	Repair and of Machinery and Equipment	5
	Information and Communication (78 Firms)	
61	Telecommunications	1
62	Computer Programming, Consultancy, and Related Activities	64
63	Information Service Activities	13
	Administrative and Support Service Activities (77 Firms)	
78	Employment Activities	1
82	Office Administrative, Office Support and Other Business Support Activities	76
	Professional, Scientific and Technical Activities (47 Firms)	
68	Real Estate Activities	1
70	Activities of Head Offices; Management Consultancy Activities	7
71	Architectural and Engineering Activities; Technical Testing and Analysis	24
72	Scientific Research and Development	5
73	Advertising and Market Research	4
74	Other Professional, Scientific and Technical Activities	6
	Transportation and Storage (15 Firms)	
52	Warehousing and Support Activities for Transportation	14
56	Food and Beverage Service Activities	1
	Utilities (14 Firms)	
38	Waste Collection, Treatment, and Disposal Activities; Materials Recovery	14
	Wholesale and Retail Trade (13 Firms)	
46	Wholesale Trade, except of Motor Vehicles and Motorcycles	13
	Education (1 Firm)	
85	Education	1

Table A2: ISCED-2013 Broad Field Classifications with Individual Major Examples

Education: Pedagogy, Specific Topics in Primary and Secondary Teaching[1]
Arts & Humanities: Music, Literature, History, Dance, Languages
Social Sciences: International Relations, Economics, Psychology, Sociology
Business Administration: Business Administration, International Commerce, Accounting
Natural Sciences, Math & Statistics: Industrial Chemistry, Biology, Geography
Information Technologies: Information Systems
Engineering: Topography, Forestry Sciences, Industrial Bioprocesses
Agriculture & Veterinary: Agronomy, Veterinary Medicine, Aquaculture
Health: Social Work, Medicine, Counseling
Tourism Services[2]: Tourist Recreation, Sustainable Tourism Business Management

Note: The individual majors are examples of those contained within each broad ISCED-2013 category and are not an exhaustive list.

[1] Social Studies, Mathematics, English, Spanish, etc.

[2] I have adapted the nomenclature of this category to better reflect the offered majors that fall into "Services".

Table A3: Mapping of Degree Fields to ISCED Broad Categories

American Community Survey Degree Category		**ISCED-2013 Broad Field Category**	
degfield	**Label**	**Category**	**Label**
11	Agriculture	8	Agriculture, Forestry, Fisheries, and Veterinary
13	Environment and Natural Resources	8	Agriculture, Forestry, Fisheries, and Veterinary
14	Architecture	7	Engineering, Manufacturing, and Construction
15	Area, Ethnic, and Civilization Studies	3	Social Sciences, Journalism, and Information
19	Communications	3	Social Sciences, Journalism, and Information
20	Communication Technologies	6	Information and Communication Technologies
21	Computer and Information Sciences	6	Information and Communication Technologies
22	Cosmetology Services and Culinary Arts	10	Services
23	Education Administration and Teaching	1	Education
24	Engineering	7	Engineering, Manufacturing, and Construction
25	Engineering Technologies	7	Engineering, Manufacturing, and Construction
26	Linguistics and Foreign Languages	2	Arts and Humanities
29	Family and Consumer Sciences	1	Education
32	Law	4	Business, Administration, and Law
33	English Language, Literature, and Composition	2	Arts and Humanities
34	Liberal Arts and Humanities	2	Arts and Humanities
35	Library Science	3	Social Sciences, Journalism, and Information
36	Biology and Life Sciences	5	Natural Sciences, Mathematics, and Statistics
37	Mathematics and Statistics	5	Natural Sciences, Mathematics, and Statistics
38	Military Technologies	7	Engineering, Manufacturing, and Construction
40	Interdisciplinary and Multi-Disciplinary Studies (General)	0	Generic
41	Physical Fitness, Parks, Recreation, and Leisure	10	Services
48	Philosophy and Religious Studies	3	Social Sciences, Journalism, and Information
49	Theology and Religious Vocations	3	Social Sciences, Journalism, and Information
50	Physical Sciences	5	Natural Sciences, Mathematics, and Statistics
51	Nuclear, Industrial Radiology, and Biological Technologies	5	Natural Sciences, Mathematics, and Statistics
52	Psychology	3	Social Sciences, Journalism, and Information
53	Criminal Justice and Fire Protection	3	Social Sciences, Journalism, and Information
54	Public Affairs, Policy, and Social Work	3	Social Sciences, Journalism, and Information
55	Social Sciences	3	Social Sciences, Journalism, and Information
56	Construction Services	7	Engineering, Manufacturing, and Construction
57	Electrical and Mechanic Repairs and Technologies	7	Engineering, Manufacturing, and Construction
58	Precision Production and Industrial Arts	7	Engineering, Manufacturing, and Construction
59	Transportation Sciences and Technologies	7	Engineering, Manufacturing, and Construction
60	Fine Arts	2	Arts and Humanities
61	Medical and Health Sciences and Services	9	Health and Welfare
62	Business	4	Business, Administration, and Law
64	History	2	Arts and Humanities

Table A4: Effect of industry exposure on field-of-study choice (Firm Count)

	STEM	Social Sciences	Arts
Baseline share	40.1%	50.8%	9.1%
Manufacturing	2.021^{***}	-1.609^{***}	-0.412^{***}
	(0.614)	(0.489)	(0.125)
Prof., Sci. and Tech. Services	-2.046^{***}	1.972^{***}	0.073^{***}
	(0.555)	(0.535)	(0.020)
Utilities	-0.695^{***}	0.508^{***}	0.187^{***}
	(0.219)	(0.160)	(0.059)
Transportation and Warehousing	-0.064^{***}	-0.336^{***}	0.399^{***}
	(0.014)	(0.074)	(0.088)
Wholesale trade	-0.496^{*}	0.583^{*}	-0.087^{*}
	(0.254)	(0.298)	(0.045)
Administrative and support services	0.269	−0.242	−0.027
	(0.242)	(0.217)	(0.024)
Information	0.149	0.013	−0.162
	(0.143)	(0.012)	(0.156)
Observations	230,034	230,034	230,034
District fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes. Entries are average marginal effects on the probability of choosing each field, in percentage points, per one within-district standard deviation increase in the industry exposure index. Estimates come from a conditional logit over three field categories with district and year fixed effects and controls for the regional unemployment rate, age, sex, secondary school type and university entrance score. The exposure index applies a Gaussian distance-decay kernel with bandwidth $h = 35$ km to counts of free trade zone firms, interacted with each industry's field-specific hiring weights.

Standard errors in parentheses are obtained by Krinsky–Robb simulation (500 draws) from a variance–covariance matrix clustered at the district level, and are rescaled by the same factor as the point estimates. Within-district standard deviations of the index are 2.788 for manufacturing and 1.560 for professional, scientific and technical services (index units).

Marginal effects sum to zero across the three categories within each industry by construction, since each student chooses exactly one field. The three estimates for a given industry are proportional to a single coefficient and therefore share one test statistic; the table reports seven independent tests, not twenty-one. Social Sciences and Professional Studies is the reference category in the underlying conditional logit, though marginal effects are defined and reported for all three categories.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A5: Effect of industry exposure on field-of-study choice (Firm log(Tenure))

	STEM	Social Sciences	Arts
Baseline share	40.1%	50.8%	9.1%
Manufacturing	3.693^{***}	-2.940^{***}	-0.752^{***}
	(0.818)	(0.651)	(0.167)
Prof., Sci. and Tech. Services	-3.297^{***}	3.179^{***}	0.118^{***}
	(0.613)	(0.591)	(0.022)
Utilities	-1.357^{***}	0.992^{***}	0.365^{***}
	(0.252)	(0.184)	(0.068)
Transportation and Warehousing	-0.073^{***}	-0.384^{***}	0.458^{***}
	(0.017)	(0.089)	(0.105)
Wholesale trade	−0.373	0.439	−0.066
	(0.384)	(0.452)	(0.068)
Administrative and support services	0.549^{*}	-0.494^{*}	-0.055^{*}
	(0.325)	(0.292)	(0.033)
Information	0.199	0.017	−0.215
	(0.164)	(0.014)	(0.178)
Observations	230,034	230,034	230,034
District fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes. Entries are average marginal effects on the probability of choosing each field, in percentage points, per one within-district standard deviation increase in the industry exposure index. Estimates come from a conditional logit over three field categories with district and year fixed effects and controls for the regional unemployment rate, age, sex, secondary school type and university entrance score. The exposure index applies a Gaussian distance-decay kernel with bandwidth $h = 35$ km to counts of free trade zone firms, interacted with each industry's field-specific hiring weights.

Standard errors in parentheses are obtained by Krinsky–Robb simulation (500 draws) from a variance–covariance matrix clustered at the district level, and are rescaled by the same factor as the point estimates. Within-district standard deviations of the index are 2.788 for manufacturing and 1.560 for professional, scientific and technical services (index units).

Marginal effects sum to zero across the three categories within each industry by construction, since each student chooses exactly one field. The three estimates for a given industry are proportional to a single coefficient and therefore share one test statistic; the table reports seven independent tests, not twenty-one. Social Sciences and Professional Studies is the reference category in the underlying conditional logit, though marginal effects are defined and reported for all three categories.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A6: Effect of industry exposure on field-of-study choice: unweighted specification (Firm Count)

	STEM	Social Sciences	Arts
Baseline share	40.1%	50.8%	9.1%
Manufacturing	3.832***	−2.473**	−1.359**
	(1.186)	(1.108)	(0.569)
Prof., Sci. and Tech. Services	−1.048	1.563**	−0.515
	(0.771)	(0.796)	(0.452)
Utilities	−0.706***	0.825***	−0.119
	(0.226)	(0.177)	(0.158)
Transportation and Warehousing	1.125**	−1.340**	0.215
	(0.543)	(0.563)	(0.332)
Wholesale trade	−0.520	2.439**	−1.919***
	(0.914)	(1.006)	(0.562)
Administrative and support services	0.189	−0.589	0.400
	(1.195)	(1.366)	(0.741)
Information	−3.578***	0.316	3.262***
	(1.312)	(1.315)	(0.738)
Observations	230,034	230,034	230,034
District fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes. Entries are average marginal effects on the probability of choosing each field, in percentage points, per one within-district standard deviation increase in the industry exposure index. This specification does *not* apply the field-industry attachment weights. The index therefore does not vary across alternatives and must enter with category-specific coefficients, which would otherwise cancel from the choice probabilities; the mapping from industries to fields is estimated rather than imposed. The specification estimates fourteen exposure parameters rather than the seven of Table A4.

Estimates come from a conditional logit over three field categories with district and year fixed effects and controls for the regional unemployment rate, age, sex, secondary school type and university entrance score. The exposure index applies a Gaussian distance-decay kernel with bandwidth $h = 35$ km to counts of free trade zone firms. Standard errors in parentheses are obtained by Krinsky–Robb simulation (500 draws) from a variance–covariance matrix clustered at the district level, and are rescaled by the same factor as the point estimates. Within-district standard deviations of the index are 2.788 for manufacturing and 1.560 for professional, scientific and technical services (index units).

Marginal effects sum to zero across the three categories within each industry by construction. Unlike Table A4, each industry here has two freely estimated coefficients rather than one, so the three category effects are not proportional to a common parameter and do not share a test statistic; significance may therefore differ across categories within a row. Social Sciences and Professional Studies is the reference category, with its coefficients normalized to zero, though marginal effects are defined and reported for all three categories.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A7: Heterogeneity in Field-of-Study Response by Sex

	Men	Women	Difference (women − men)
Baseline share of students (%)			
STEM and Applied Sciences	50.4	32.5	−17.9
Social Sciences and Prof. Studies	41.8	57.5	15.7
Arts, Writing and Culture	7.8	10.1	2.2
Panel A. Manufacturing			
STEM and Applied Sciences	1.634**	2.305***	0.671***
	(0.656)	(0.576)	(0.128)
Social Sciences and Prof. Studies	−1.308**	−1.828***	−0.520***
	(0.525)	(0.457)	(0.105)
Arts, Writing and Culture	−0.326**	−0.477***	−0.151***
	(0.131)	(0.119)	(0.024)
Response ratio $\hat{\beta}_F/\hat{\beta}_M$		1.616	
Panel B. Professional, Scientific and Technical Services			
STEM and Applied Sciences	−2.783***	−1.424***	1.359***
	(0.615)	(0.533)	(0.130)
Social Sciences and Prof. Studies	2.601***	1.408***	−1.193***
	(0.575)	(0.528)	(0.107)
Arts, Writing and Culture	0.182***	0.015***	−0.167***
	(0.040)	(0.006)	(0.035)
Response ratio $\hat{\beta}_F/\hat{\beta}_M$		0.583	
Observations	230,034	230,034	230,034
District fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes

Notes. Entries are average marginal effects on the probability of choosing each field, in percentage points, per one within-district standard deviation of the industry exposure index.

The Difference column is the heterogeneity test. It is a linear contrast of the estimated coefficient vector, evaluated using the full district-clustered covariance matrix. Because the men's and women's marginal effects are both functions of the same exposure coefficient, they are strongly positively correlated ($\hat{\rho} = 0.99$), and the contrast is estimated far more precisely than either level. Overlap between the men's and women's confidence intervals therefore carries no implication for the significance of their difference (**?**).

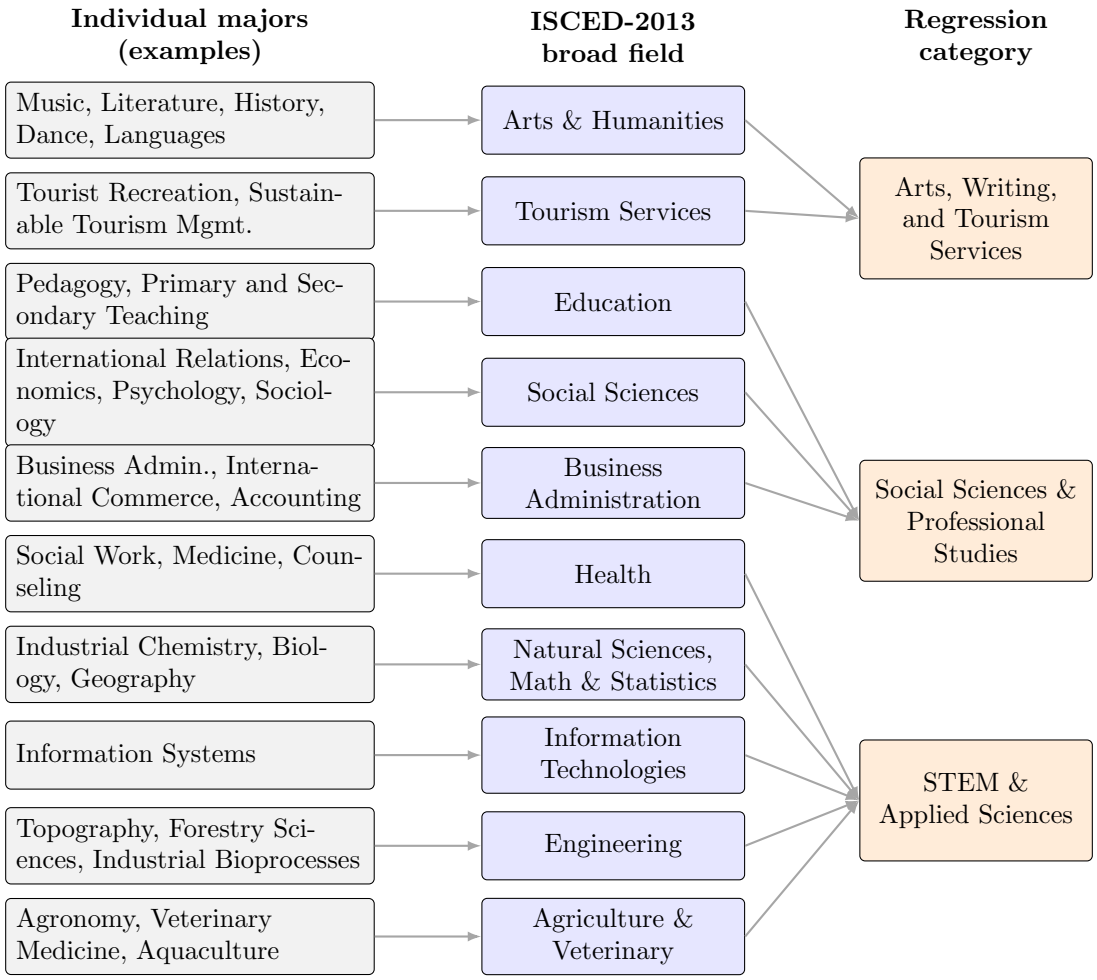
The response ratio $\hat{\beta}_F/\hat{\beta}_M$ is recovered from the interaction coefficient and is free of baseline-composition effects. Because women's baseline STEM share is 17.9 percentage points below men's, and marginal effects scale with the baseline probability, the difference in marginal effects understates the underlying behavioural difference for manufacturing and overstates it for professional services.

Marginal effects sum to zero across categories within each panel and sex.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

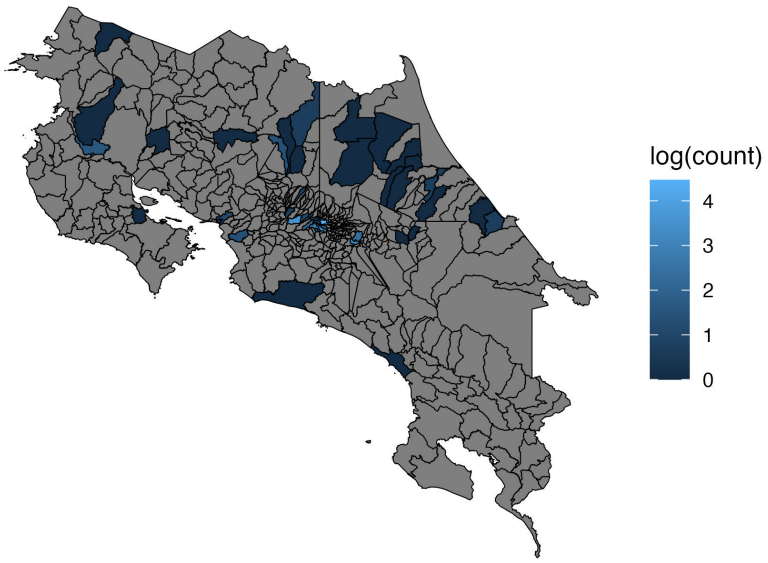
Appendix B - Figures

Figure A1: From individual majors to ISCED-2013 broad fields to the three major categories used in the estimation

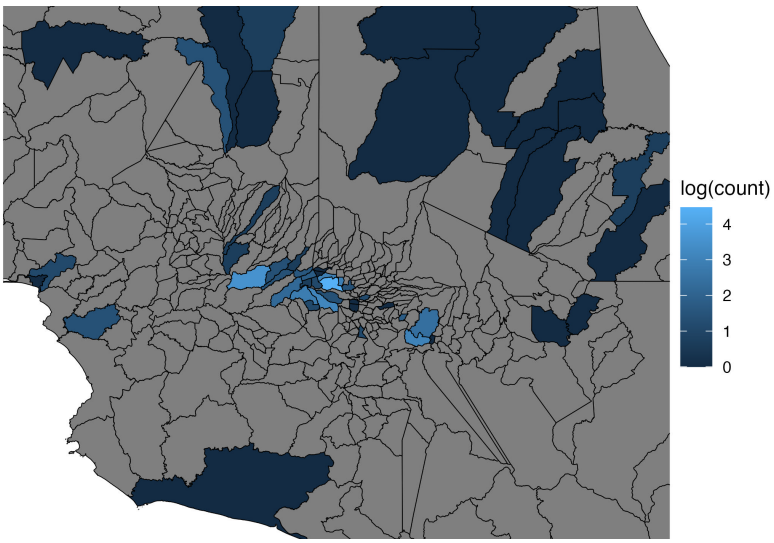


Note: The figure illustrates how individual degrees are aggregated into the three categories used in the main estimation. The left column contains only a subset of majors as examples and should not be considered as an exhaustive list. Majors are first assigned to broad fields following the UNESCO International Standard Classification of Education, Fields of Education and Training (ISCED-F 2013), using the classification supplied by CONARE, the constitutional coordinating body for Costa Rica's public higher education system. Broad fields are then collapsed into three categories, aggregated via similarities amongst them. The same three-category partition defines the industry–field attachment weights p_{mjt} used to construct the exposure measure .

Figure A2: Spatial Distribution of Free Trade Zone Firms



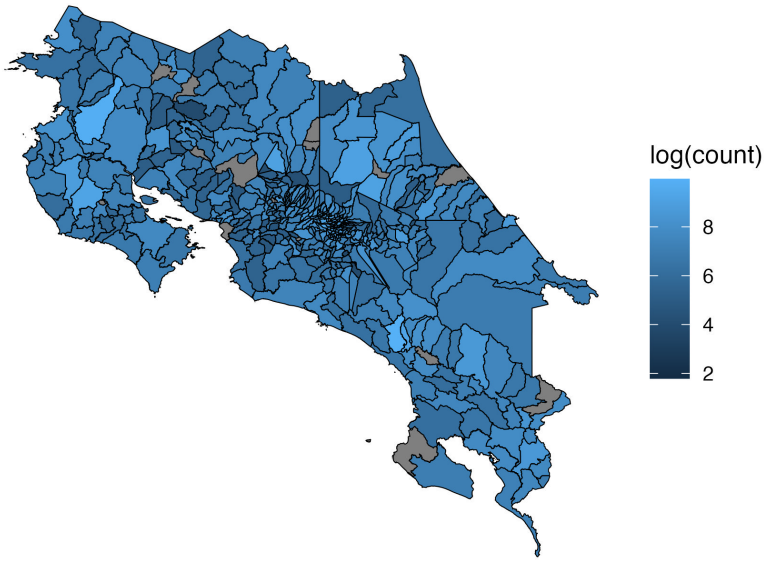
(a) National



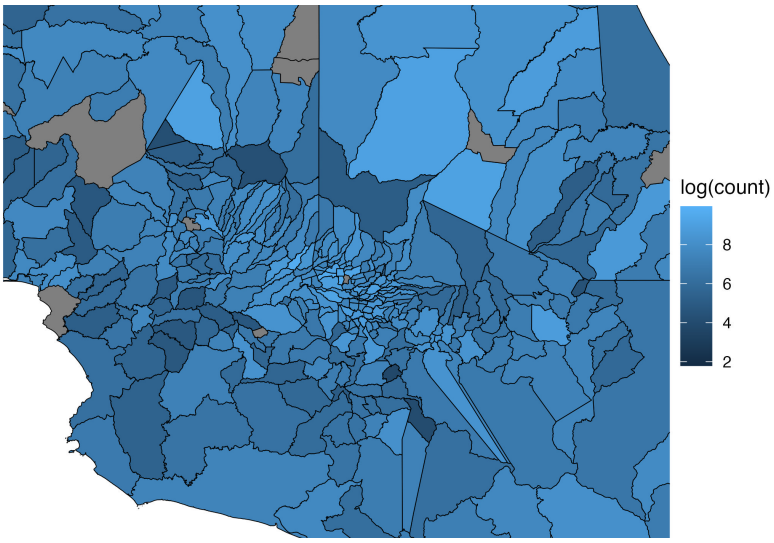
(b) Greater Metropolitan Area

Note: Each polygon is a district, the geographic unit at which exposure is measured. Shading gives the natural log of the number of free trade zone firms located in the district as of the end of the sample period. Districts in grey contain no FTZ firms and are therefore undefined. Firms are heavily concentrated in the GAM, with scattered presence along the Pacific corridor and near the Caribbean port. All industries are pooled here, the estimation treats the seven industries separately. Panel A2a shows the entire nation, while panel A2b is a detail of the central region. Sources: Firm locations are sourced from PROCOMER and district boundaries are from the Database of Global Administrative Areas (GADM), version 4.1, administrative level 3 (site).

Figure A3: Spatial Distribution of Student Applications



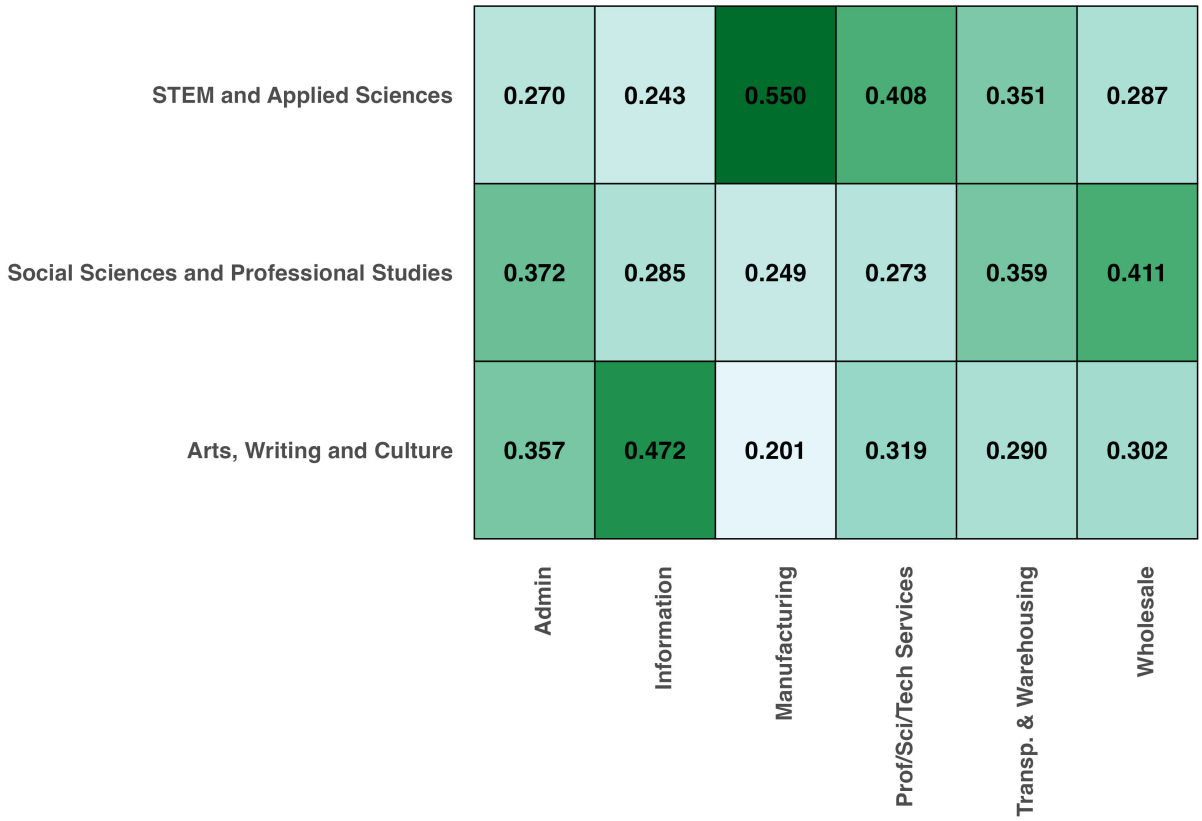
(a) National



(b) Greater Metropolitan Area

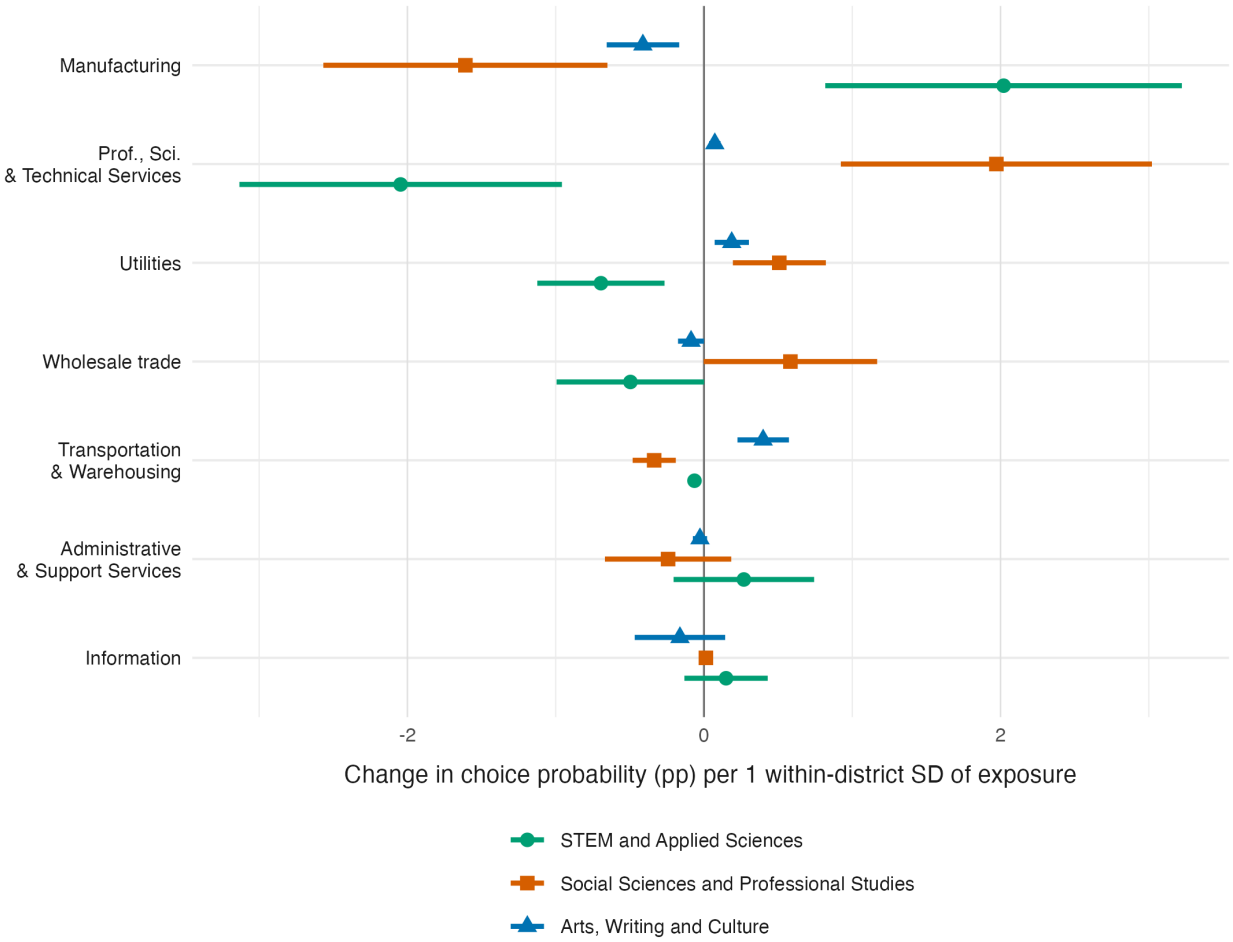
Note: Each polygon is a district, the geographic unit at which exposure is measured. Shading gives the natural log of the number of student applicants located in the district aggregated across the sample period. The image shows that students submit applications from all across the nation. All fields choices are pooled here, the estimation treats choices individually. Panel A3a shows the entire nation, while panel A3b is a detail of the central region. Sources: Student locations are sourced directly from the student application data provided by UNA and district boundaries are from the Database of Global Administrative Areas (GADM), version 4.1, administrative level 3 (site).

Figure A4: Within Sample Distribution of Probability Attachment Weights Between Field of Study and Industry



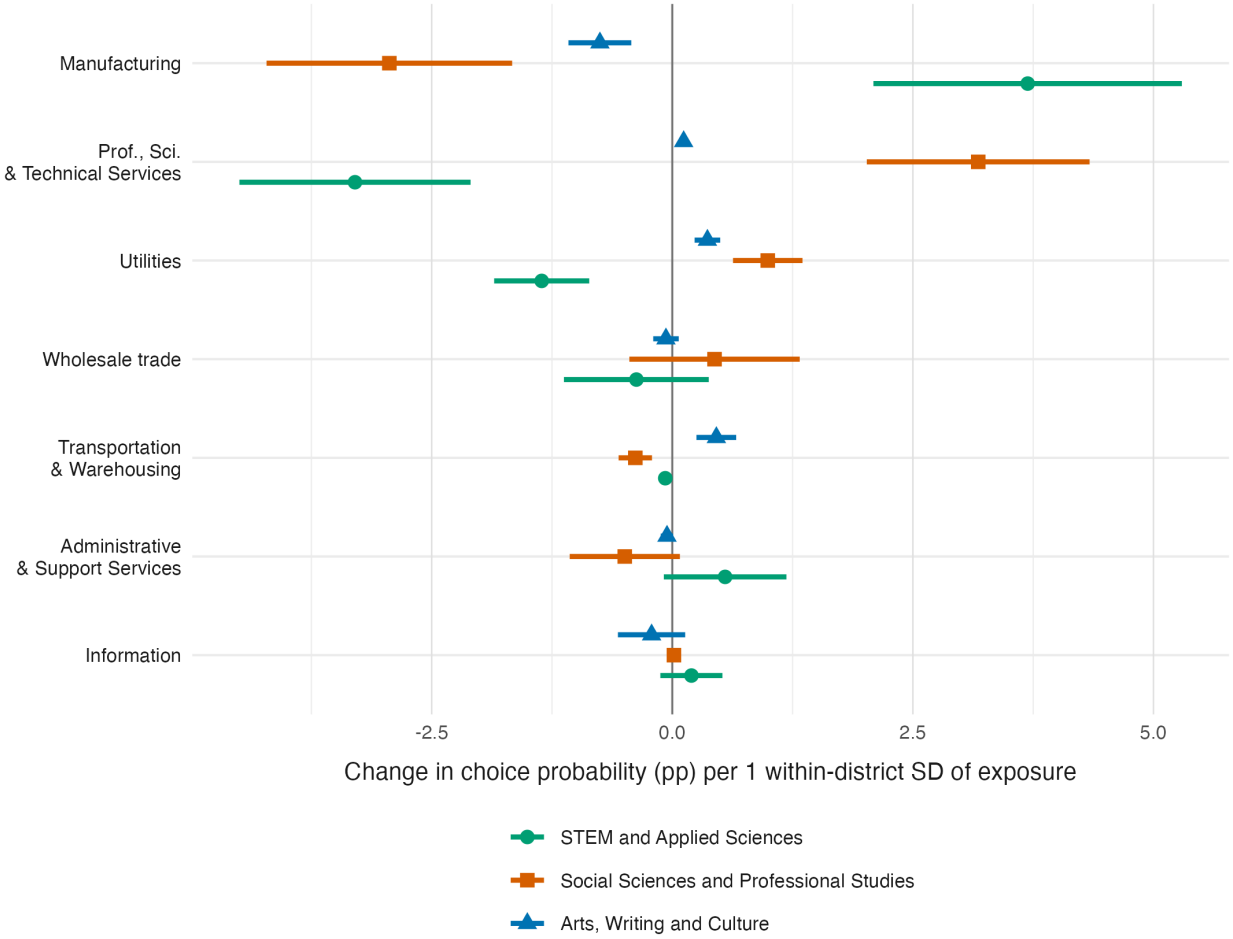
Note: Each cell contains the aggregated probability attachment weight assigned to the respective field of study (vertical axis) by the industry (horizontal axis) across the sample period. Each industry column sums up to 1, representing the total probability of employment within-industry. Each cell is shaded relative to their value, with darker colors representing a higher attachment weight. Sources: American Community Survey responses for the sample period.

Figure A5: Average Marginal Effects for Firm Count Specification



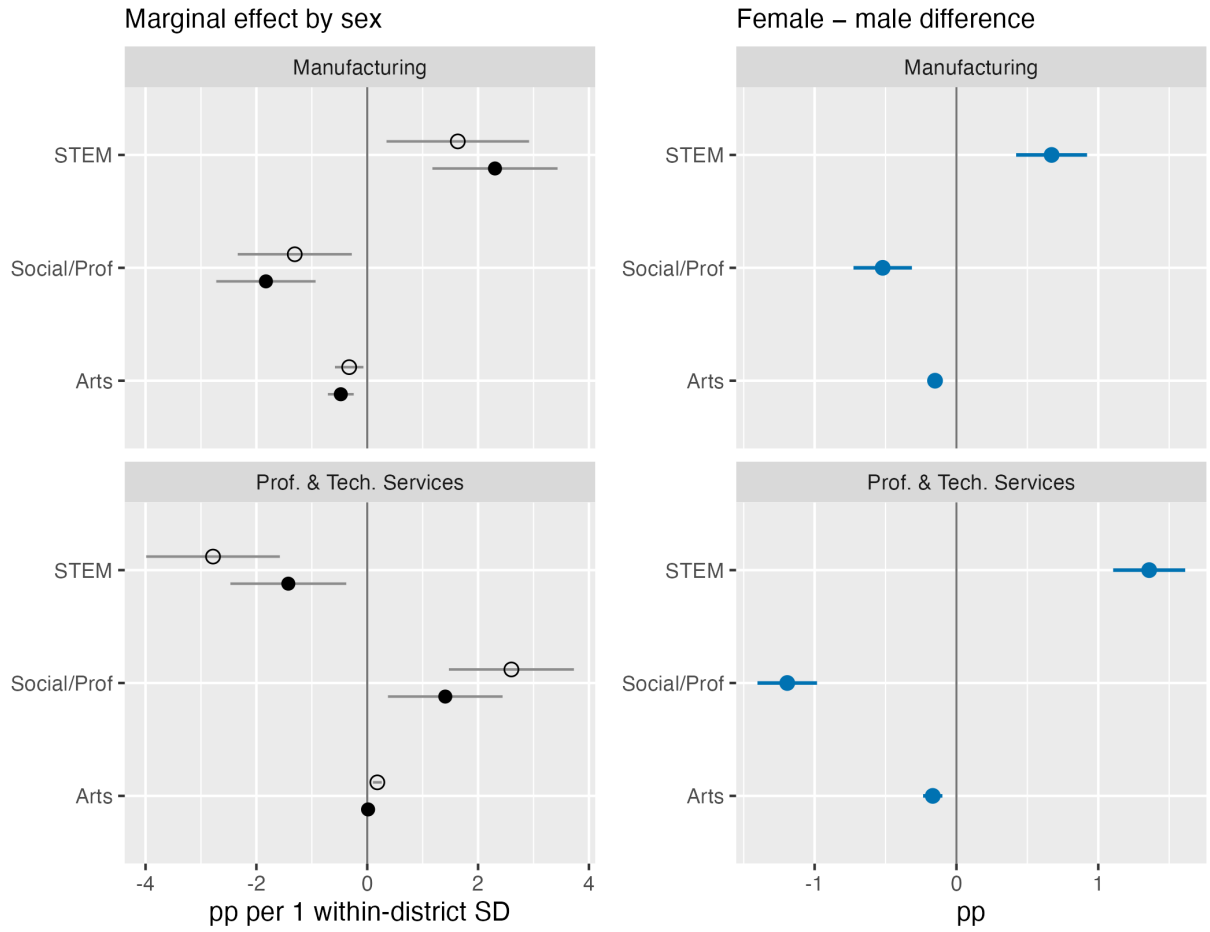
Note: Points are average marginal effects on the probability of choosing each field, in percentage points, per one within-district standard deviation increase in industry exposure; horizontal bars are 95% confidence intervals. Estimates come from a conditional logit over the three field categories with district and year fixed effects, interacted with category, and controls for the regional unemployment rate, age, sex, secondary school type of the applicant and university entrance score. Exposure applies a Gaussian distance-decay kernel with bandwidth $h = 35$ km to counts of free trade zone firms, weighted by each industry's field-specific hiring shares. Confidence intervals are obtained by Krinsky–Robb simulation (500 draws) from a district-clustered covariance matrix. Within each industry the three effects sum to zero by construction, since each student chooses exactly one field; the three estimates are proportional to a single coefficient and share one test statistic, so the figure displays seven independent tests rather than twenty-one. Manufacturing is the only industry that shifts students toward STEM; professional services, utilities and wholesale trade move them in the opposite direction, toward Social Sciences and Professional Studies. Within-district standard deviations of the index are reported in Table [X]

Figure A6: Average Marginal Effects for Log Tenure Specification



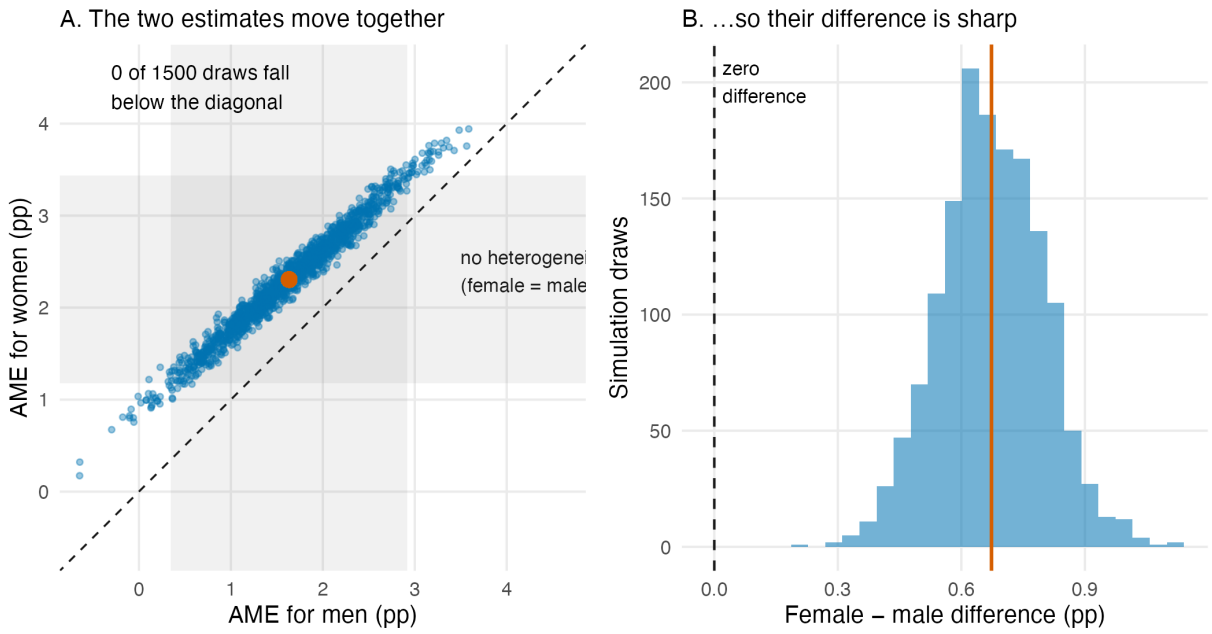
Note: Points are average marginal effects on the probability of choosing each field, in percentage points, per one within-district standard deviation increase in industry exposure; horizontal bars are 95% confidence intervals. Estimates come from a conditional logit over the three field categories with district and year fixed effects, interacted with category, and controls for the regional unemployment rate, age, sex, secondary school type of the applicant and university entrance score. Exposure applies a Gaussian distance-decay kernel with bandwidth $h = 35$ km to the natural log of tenure of free trade zone firms, weighted by each industry's field-specific hiring shares. Confidence intervals are obtained by Krinsky–Robb simulation (500 draws) from a district-clustered covariance matrix. Within each industry the three effects sum to zero by construction, since each student chooses exactly one field; the three estimates are proportional to a single coefficient and share one test statistic, so the figure displays seven independent tests rather than twenty-one. Manufacturing is the only industry that shifts students toward STEM; professional services, utilities and wholesale trade move them in the opposite direction, toward Social Sciences and Professional Studies. Within-district standard deviations of the index are reported in Table [X]

Figure A7: Heterogeneity Between Male and Female for Manufacturing and Professional/Technical Services



Note: Left panels show the average marginal effect of industry exposure on the probability of choosing each field, separately for men (hollow) and women (solid), in percentage points per one within-district standard deviation of exposure. Right panels show the female-minus-male difference. Bars are 95% confidence intervals. Each industry's estimates come from a single conditional logit in which that industry's exposure is interacted with an indicator for female, with district and year fixed effects and the full set of individual controls; the remaining industries enter without interactions. The right panel is the heterogeneity test. The difference is a linear contrast evaluated using the full district-clustered covariance matrix. Because the male and female marginal effects are both functions of the same exposure coefficient, they are strongly positively correlated ($\hat{\rho} = 0.99$) and their difference is estimated far more precisely than either level; overlap between the male and female intervals in the left panels therefore carries no implication for the significance of the difference (Schenker and Gentleman, 2001). Because women's baseline STEM share is 17.9 percentage points below men's, and marginal effects scale with baseline probability, the differences understate the underlying behavioural gap for manufacturing and overstate it for professional services. Within each panel the three categories share one test statistic.

Figure A8: Heterogeneity Between Male and Female for Manufacturing



Note: The figure illustrates why the overlapping confidence intervals in Figure A7 are compatible with a precisely estimated difference. It plots the 1,500 Krinsky–Robb simulation draws of the average marginal effect of manufacturing exposure on the probability of choosing STEM, for men (horizontal axis) against women (vertical axis). The orange point marks the estimates. Shaded bands are the marginal 95% confidence intervals for each subgroup; they overlap across most of their length. The dashed line is the 45-degree line, along which the two effects would be equal. Because both effects are functions of the same exposure coefficient, the draws are strongly correlated ($\hat{\rho} = 0.99$) and lie in a narrow band parallel to the 45-degree line: uncertainty concerns how large the manufacturing effect is, not which group responds more strongly. No draw falls below the line. Panel B shows the implied distribution of the female-minus-male difference, which is centred well away from zero. Overlap between marginal confidence intervals is therefore uninformative about the significance of their difference.